

DATA 202 — Quiz 2

Week 4 — Covering Week 3: Cleaning, Grouping & Plots

Name: _____

Date: _____

The Dataset: People Experiencing Homelessness

`homeless` contains 100 simulated records of people experiencing homelessness, cleaned the same way as in Monday's class: standardized city names, shelter status, and education levels.

Columns: `id`, `years_homeless`, `family_size`, `monthly_support_usd` (numerical) · `name`, `city`, `shelter_status`, `education_level` (categorical/text)

First few rows (already cleaned):

id	name	city	shelter_status	years_homeless	family_size	monthly_support_usd	education_level
2	Joseph	Boston	street	10	2	366	None
8	Lucas	San Francisco	shelter temporary	8	2	25	Primary
21	Lucas	Boston	unsheltered	9	5	84	None
39	Anna	Los Angeles	shelter	12	5	556	Primary
59	David	San Francisco	shelter	11	2	264	Secondary

Question 1 (SLO 03A)

The raw (uncleaned) `shelter_status` column also has spacing variants like `"shelter , pending"`, `"shelter,pending"`, and `"shelter , pending"` — the same status, just spaced differently. Complete the code below (which already lowercases the text) to standardize all of these into `"shelter pending"`.

```
homeless["shelter_status"] = (  
    homeless["shelter_status"]  
    .str.strip()  
    .str.lower()  
    .str.replace(r"_____", "shelter pending", regex=True)  
)
```

Question 2 (SLO 03B)

Complete the code below to compute, for each `education_level`, both the number of people and the average `years_homeless`, using named aggregation.

Then answer: if you used `"median"` instead of `"mean"` for `avg_years`, would the result be more or less sensitive to one extremely high outlier in `years_homeless`?

```
result = homeless.groupby("_____").agg(
    n_people=("id", "count"),
    avg_years=("_____", "_____"),
)
```

Question 3 (SLO 03C)

For each question below about the `homeless` data, name the best plot type (Histogram / Scatter / Line / Bar) and which column(s) you'd map to x (and y, if relevant).

Question	Plot Type	Column(s)
What's the distribution of <code>years_homeless</code> across all people?		
How do <code>family_size</code> and <code>monthly_support_usd</code> relate to each other?		
How does average <code>family_size</code> change as <code>years_homeless</code> increases?		
How does average <code>years_homeless</code> compare across <code>shelter_status</code> values?		

Question 4 (SLO 02A) — Review from Quiz 1

For each command below, identify whether it **Accesses**, **Adds**, or **Deletes** data, and whether it operates on a **Row** or a **Column**.

#	Command	Access / Add / Delete?	Row(s) or Column?
1	<code>homeless["years_homeless"]</code>		
2	<code>homeless["long_term"] = homeless["years_homeless"] > 10</code>		
3	<code>homeless = pd.concat([homeless, pd.DataFrame([new_row])], ignore_index=True)</code>		
4	<code>homeless = homeless[homeless["city"] != "Boston"]</code>		
5	<code>homeless = homeless.drop(columns=["family_size"])</code>		

Question 5 (SLO 02B) — Review from Quiz 1

Complete the code below to select all people in “**San Francisco**” with `monthly_support_usd` below 100, sorted by `monthly_support_usd` from lowest to highest.

Remember: the filter condition still goes inside the square brackets and must reference `homeless` again, e.g. `homeless[homeless["Column"] ...]`.

```
result = homeless[
    _____
]
result = result.sort_values("_____", ascending=_____)
print(len(result))
```

Question 6 (SLO 02C) — Review from Quiz 1

Available visual channels: `x`, `y`, `color`, `size`, `symbol`, `facet_col`.

As on Quiz 1: since this is a paper quiz, `color` isn't something you can sketch in pencil — the code below uses `symbol` instead, mapping a categorical column to different *marker shapes* (e.g. `×`, `○`, `△`) rather than colors.

For the code below, fill in the table: for each visual channel it uses, name the column mapped to it and say whether that column is Numerical or Categorical.

```
px.scatter(homeless, x="family_size", y="monthly_support_usd",
           symbol="education_level", size="years_homeless",
           title="Family Size vs. Monthly Support")
```

Visual Channel	Column Used	Data Type (Numerical / Categorical)
x-axis		
y-axis		
symbol		
size		

In the space below, sketch what you expect this chart to look like. Label your axes. You don't need exact data — just show the general pattern (point cloud, marker shapes (`×`/`○`/`△`) for groups, varying point sizes).